

MLFF ADAPT-PREC1 binary model precision specification

Status: implemented in mdstats 0.20.122a0.

Purpose

ADAPT-PREC1 defines one precision variable for the learned MACE model and removes staged FP32-to-FP64 refinement from new production campaign identity. It supersedes the production precision semantics of PREC1-PREC3 without rewriting historical campaign evidence.

Canonical model modes

New campaign initialization supports exactly:

- **single**: learned-model parameters/buffers, training/autograd, optimizer/EMA state, and every MACE inference path use float32;
- **double**: the same learned-model lifecycle uses float64.

Plain `init` resolves to **single**. **refine** and user-facing **mixed** are not production model modes. No hardware heuristic may change the selected model dtype.

The selected dtype must agree across `[model].dtype`, `[training].dtype`, `[evaluation].dtype`, `[verification].dtype`, and `[export].dtype`. Production commands fail closed on a mismatch.

Inference and export invariant

A checkpoint is evaluated, verified, used by committee/member inference, and exported in its learned-model dtype. An FP32 checkpoint must not be cast to FP64 merely to satisfy a separate evaluation or deployment dtype. Checkpoint/template reconstruction therefore forbids model-dtype promotion in production evaluation.

FP64 scientific-arithmetic invariant

Model dtype does not control mdstats-owned numerical analysis. Operations that are cheap relative to model inference and benefit from wider arithmetic remain float64 under both model modes, including:

- elemental/reference-energy fitting, rank/SVD diagnostics, PCA/QR/SVD, feature scaling, and deterministic selection-distance reductions;
- cell/strain/geometric linear algebra;
- SSE/RMSE/MAE, regression, confidence/statistical aggregation, and checkpoint scoring;
- observable analysis and NVE drift fitting; and

- mdstats-owned persistent MD positions, cell, velocities/momenta, and integration bookkeeping where applicable.

The qualified critical energy/virial reduction policy is also FP64 where that adapter is applicable. This does not make an FP32 network an FP64 model; operations inseparable from the MACE differentiation graph remain in the selected learned-model dtype.

New-campaign protocol identity

New campaign TOML contains no `[training.precision]` staged schedule. New optimizer and DATA8 protocol identities therefore carry no resolved precision schedule and cannot reach the historical PREC2 optimizer/EMA promotion runtime. Binary model dtype is bound through the ordinary model/training/evaluation/verification/export identities.

Historical compatibility

Historical staged `refine` schedules remain deserializable for reporting, storage, audit, and archive operations. They are read-only historical evidence. Production `prepare`, `preflight`, `train`, `evaluate`, and `verify` fail closed when such a profile is supplied and require the user to choose a new `single` or `double` scientific identity. Historical schedules are never silently flattened or reinterpreted.

Acceptance tests

The gate is qualified only when tests prove:

1. CLI initialization exposes only `single|double` and defaults to `single`;
2. generated configs contain no staged precision schedule;
3. model dtype is identical across training, evaluation, verification, and export;
4. evaluation forbids checkpoint/template dtype promotion;
5. FP64 critical/scientific arithmetic remains active for both learned-model dtypes;
6. new DATA8/optimizer policies are schedule-free; and
7. historical `refine` evidence remains readable while production execution rejects it.